

Project Proposal: Addressing Wearable Sleep Tracking Inequity: A New Dataset and Novel Methods for a Population with Sleep Disorders

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Introduction

Sleep is one of the most fundamental pillars of human health, but the gold standard for assessing sleep quality (Polysomnography), is expensive, labor-intensive, and requires patients to sleep in controlled laboratories for monitoring. These artificial sleep settings don't capture natural sleeping patterns that are needed to diagnose chronic sleeping disorders. While consumer wearables have emerged as a scalable alternative for clinical monitoring, their development has relied on data from younger and healthier populations. This has created an inequity in digital health, where existing sleep tracking algorithms fail when applied to individuals that don't match the training data distribution (e.g., individuals with sleep disorders or irregular sleep patterns).

A key challenge is that clinical sleep datasets are both small and heterogenous. Collecting sleep quality related wearable data is expensive and very labor intensive, which limits the size of available datasets and makes it difficult for models to learn robust patterns in the training data signals. At the same time, measurements such as heart rate and movement vary between individuals due to demographic and physiological differences, such as age, medications, and underlying health conditions.

Additionally, sleep cycles are sequential, with nightly transitions between stages. Standard approaches don't capture these temporal dependencies when trained on small datasets. Together, these challenges create models which generalize very poorly to populations with sleep disorders, resulting in unreliable monitoring via wearables for the populations that need it most.

Methodology

Specific Approach

Wang et al. (2024) address wearable-based sleep stage classification for patients with sleep disorders using the DREAMT dataset — 100 overnight recordings from the Duke Sleep Disorders Center, pairing Empatica E4 signals (BVP, ACC, EDA, TEMP, HR, IBI) with PSG-annotated sleep stage labels. The pipeline has two stages. First, an epoch-level classifier processes hand-engineered features

extracted from 30-second windows — including HRV metrics, EDA decomposition, and Gaussian-smoothed temporal derivatives — fed into a mixed-effects GPBoost model that incorporates either obesity or apnea severity as a random effect to account for inter-subject physiological heterogeneity. Second, a shallow single-layer unidirectional LSTM post-processor takes the GPBoost output probabilities and the two most predictive features (ACC INDEX and HRV HFD) as input, performing sequence-to-sequence classification to recover temporal sleep-state dynamics. Performance is evaluated via 5-fold participant-level cross-validation using F1, AUROC, AUPRC, accuracy, and Cohen's Kappa.

For the PyHealth contribution, we will implement a standalone `SleepWakeTask` for the existing DREAMT dataset in PyHealth. This task will define the input feature schema (per-epoch E4-derived features), the binary wake/sleep label extraction from PSG annotations, the 30-second epoch windowing logic, and the participant-level train/validation/test split interface consistent with PyHealth conventions.

Novelty, Relevance, and Hypotheses

The paper makes two distinct methodological contributions. First, it is the first work to apply Gaussian Process Boosted Trees (GPBoost) to wearable sleep tracking, using clinically meaningful group variables (obesity, apnea severity) as random effects. This directly addresses a known failure mode of existing models: treating a heterogeneous sleep-disordered population as homogeneous, which degrades performance. Second, the LSTM post-processing stage is novel in this context — prior work relied on Hidden Markov Models or rule-based smoothing (Wang et al. 2024), neither of which captures long-range temporal dependencies spanning the full overnight window.

Together, the authors hypothesize that (1) mixed-effects modeling improves generalization across physiological subgroups, and (2) LSTM post-processing recovers temporal structure that epoch-level classifiers inherently discard. Both hypotheses are supported by the ablation results in Table 2 of the original paper: each component contributes incrementally, with the full model (GPBoost + Apnea Severity + LSTM) achieving the best results (F1: 0.823, AUROC: 0.926, Cohen's Kappa: 0.695).

Ablations and Extensions Planned

We propose two ablations, both novel relative to what is reported in the paper and implementable through the `SleepWakeTask` interface.

Ablation 1 — Signal subset ablation. The paper uses all available E4 signals jointly but does not report the contribution of individual modalities. We will vary the input feature set across clinically motivated subsets — ACC-only, BVP/HRV-only, EDA+TEMP-only, and all signals combined — to quantify which modalities drive performance. This has direct practical relevance for deployment on lower-cost devices that may not support all sensors.

Ablation 2 — Task label granularity. Starting from the binary `SleepWakeTask`, we will extend to a multi-class `SleepStageTask` (Wake / REM / N1 / N2 / N3) to test whether the pipeline generalizes beyond the paper’s binary framing. We hypothesize performance will degrade most on N1 — the most ambiguous stage — and that the LSTM post-processor’s learned wake-onset and wake-offset patterns will not transfer cleanly to the more complex transition structure of full sleep staging.

Data Access and Implementation Details

Dataset and Access

The proposed project uses the DREAMT dataset introduced by Wang et al. (2024). The dataset contains approximately 100 overnight sleep recordings collected at the Duke Sleep Disorders Center. Each recording includes synchronized wearable signals from the Empatica E4 device, including blood volume pulse (BVP), accelerometer (ACC), electrodermal activity (EDA), temperature (TEMP), heart rate (HR), and inter-beat interval (IBI). These wearable measurements are paired with ground-truth sleep stage annotations obtained through clinical polysomnography (PSG), which serves as the gold standard for sleep monitoring.

Unlike many wearable sleep datasets that focus on healthy populations, DREAMT contains recordings from patients with diagnosed sleep disorders. This makes the dataset particularly relevant for evaluating wearable-based sleep tracking models in clinical populations where physiological signals and sleep patterns differ significantly from those of healthy individuals. The dataset has been integrated into the PyHealth framework, allowing it to be accessed directly through PyHealth’s standardized dataset interface.

Implementation in PyHealth

Because the DREAMT dataset is already implemented within PyHealth, our work will focus on defining a task-specific interface rather than recreating the dataset itself. Specifically, we will implement a `SleepWakeTask` that operates on the DREAMT dataset.

Each overnight recording will be segmented into 30-second epochs consistent with the annotation structure used in the original study. Wearable-derived features computed from the Empatica E4 signals will serve as model inputs, while PSG-derived annotations will be used to generate binary sleep/wake labels. The task interface will also define

participant-level train, validation, and test splits to prevent subject-level data leakage.

Since the recordings represent sequential overnight signals, the data will be exposed as ordered time-series inputs compatible with temporal models such as recurrent neural networks.

Code Usage

We will reuse the existing DREAMT dataset implementation provided in PyHealth. However, the modeling pipeline described in the paper will be implemented independently. In particular, we will reproduce the two-stage approach proposed by Wang et al. (2024), consisting of a GPBoost-based epoch-level classifier followed by an LSTM temporal post-processing model. Implementing the model within PyHealth ensures compatibility with the framework’s dataset abstractions and training utilities while allowing us to extend the original approach with additional ablations and experiments.

References

Wang, W. K.; Yang, J.; Hershkovich, L.; Jeong, H.; Chen, B.; Singh, K.; Roghanizad, A. R.; Shandhi, M. M. H.; Spector, A. R.; and Dunn, J. 2024. Addressing Wearable Sleep Tracking Inequity: A New Dataset and Novel Methods for a Population with Sleep Disorders. In *Proceedings of the Conference on Health, Inference, and Learning (CHIL)*, volume 248, 380–396. PMLR.